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India's Space Economy, 2011–12 to 2020–21: Its Size and Structure

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ABSTRACT

India is one of the few developing countries with a growing technological capability in the space industry. This capability manifests itself in designing, manufacturing and launching satellites of various sizes and sophistication. The state has played a significant role in evolving and nurturing this technological capability, as the space sector has received considerable public investment. The space economy has grown and can be essential to the country's high-tech manufacturing and services sectors. A necessary feature of this technological capability building is entirely built through indigenous research and development by the Indian Space Research Organisation. Employing an eclectic framework, the paper maps out the size and structure of India's space sector and analyses the state's role in shaping it. It also provides us with some estimates of the productivity of those public investments in creating a space economy.

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1. Introduction

India is ranked as one of the major space powers in the world because of its acknowledged capability along the full range of the space technology spectrum and substantial indigenous capability. The Indian space sector has over five decades of successful existence. Four characteristics of the space sector in India stand out. First, it is one of the country's rare examples of domestic technological capabilities purely through indigenous efforts and in a high-techno sector. Second, these efforts were spearheaded entirely by a public sector organisation. Third, initially, the state has collaborated with a certain limited number of private sector enterprises in building up this technological capability, and of late, the role of the market in this arena has been enlarged. Fourth, it illustrates the country as a hub for generating frugal innovation in space products and services. India can manufacture and launch satellites of various sizes at a competitive cost in comparison to international price, while more evident for launches, it has demonstrated to satellite manufacture also. However, the space sector in India has evolved into an important industry with a rapidly changing manufacturing component. One of the most recent developments in this area is the emergence of some startups, some even developing the technological capability to design and manufacture small satellites and launch vehicles. The space economy of India is in a state of flux, where through recent policy changes, the government has sought to reduce its prominent position and is allowing the private sector also to play a vital role not only just in the manufacturing of satellite launch vehicles but also in the provision of communication and remote sensing services.

Further, with the growth of the digital industries, the requirement for products and services from the space sector is bound to increase tremendously. However, notwithstanding its existence

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over 50 years or so, there is very little understanding of the contours of this industry. In this context, the main objective of this paper is to map out the size of India's space economy. Such a mapping will allow us to understand the impact of the space economy on the rest of India's economy.

The paper is structured into five sections. Section 2 surveys the available methodologies to measure the size of the space economy. Section 3 deals with the literature on India's space sector. Section 4 describes the contours of the space economy in terms of the organization for economic cooperation and development (OECD) framework. Section 5 measures the size and structure of the economy and traces its trends over the years. Section 6 concludes with the study's policy implications.

2. Analytical frameworks for estimating the size of the space economy

Although several countries are engaged in space-related activities, all of the estimates available concern the U.S. space economy. There are essentially three significant estimates of the space economy. These are the Satellite Industry Association's (S.I.A.'s) annual *State of the Satellite Industry Report*, the Space Foundation's *The Space Report* and the Organisation for Economic Cooperation and Development's *The Space Economy in Figures: How Space Contributes to the Global Economy* provides an initial overview of the global space sector. There have been four such reports published in 2007, 2011, 2014 and 2019. But the most important one dealing very specifically with the methodological challenges in measuring the size of the space economy is by OECD (2012) [23]. A second handbook outlining the scope of the space economy is under preparation. However, the following working definition is used extensively worldwide (OECD, 2019) [25].

> "*The space economy is the full range of activities and the use of resources that create and provide value and benefits to human beings in the course of exploring, understanding, managing and utilising Space. Hence, it includes all public and private actors involved in developing, providing and using space-related products and services, ranging from research and development, the manufacture and use of space infrastructure (ground stations, launch vehicles and satellites) to space-enabled applications (navigation equipment, satellite phones, meteorological services, etc.) and the scientific knowledge generated by such activities. It follows that the space economy goes well beyond the space sector itself since it also comprises the increasingly pervasive and continually changing impacts (both quantitative and qualitative) of space-derived products, services and knowledge on economy and society*".

We will first elaborate on the OECD framework and then provide a critical review.

2.1. The OECD framework

According to this framework, there are three broad ways to conceive the size of the space economy of any country. The three indicators are as follows:

- Inputs to the space economy
- Activities and outputs of the space economy
- Impacts of the space economy

(i) **Inputs to the space economy**: This is usually measured by three separate indicators such as (a) share of budget for space activities in a country's gross domestic product (GDP); (b) assets of the space sector in terms of assets in orbit and on the ground; (c) human capital employed in the space sector consisting of engineers, scientists and marketing specialists. The majority of the people working in the sector have a science, mathematics, engineering or information technology background.

(ii) **Activities and outputs of the space economy**:

- **The manufacturing space industry is measured by the revenues and employment generated by the construction of satellites and launchers** and their associated services. The space manufacturing sector is relatively speaking a small industry. Several interesting space startups have appeared in the space manufacturing sector.
- **The satellite telecommunications sector** is by far the most critical space-related commercial market. Much of the satellite communications business comes from television-direct-to-home (D.T.H.) satellite dishes, direct broadcast satellite, mobile telephony, broadband services, and so on.
- **The satellite earth observation sector** is the remote sensing sector. It allows the measurements from the orbit of an extensive range of geophysical parameters, spanning the whole spectrum of the environment, including the atmosphere, land, oceans, ice and snow. Commercial satellite earth observation is a niche market. Launches of Earth-Observation Remote Sensing and Meteorology Satellites and their revenue generation can be estimated.
- **The insurance market for space activities**—A branch of the insurance sector that explicitly covers the operations of the commercial space sector. The main risks covered still tend to be a failure at launch or mechanical troubles for telecommunications satellites.
- **International trade in selected space products**—Exports and imports of spacecraft and spacecraft launch vehicles. France, the United States, Belgium, Italy and Germany lead spacecraft exports (including satellites) and spacecraft launch vehicles.
- **Innovation for future economic growth**—Analysis of patents provides some insight into innovative activities concerning the electrical and mechanical machinery and equipment required for space-based systems (satellites, launchers) and downstream applications, such as telecommunications navigation systems. The countries' share in space-related patents over the 2000–08 period shows the United States and Europe leading, followed by Korea and Japan. The Russian Federation, France, Israel and the United States show much patenting in space activities, compared to other economic sectors.
- **Space launch activities worldwide**—Ten countries have demonstrated independent orbital launch capabilities, and seven countries (i.e. the United States, the Russian Federation, China, Japan, India, Israel and Iran) and the European Space Agency have operational launchers.
- **Space exploration activities**—Space exploration remains a crucial driver for investments in innovation and sciences, and it constitutes an intensive activity for significant space agencies and industries. Considerable achievements have attracted great public interest (e.g. landing on the Moon; Mars exploration by robots; probe landing on Titan).

The OECD framework helps delineate the contours of the space economy within a nation. But the framework will not result in a summary measure of it as expressed as a per cent of its G.D.P. This is because the input and output indicators contained in the

framework are a motley assortment of physical and value-based numbers. Apart from the OCED, the size of the world space economy has been estimated by three more agencies, namely the Space Foundation, the S.I.A. and the institute for defence analyses (IDA) Science and Technology Policy Institute. Of these, the I.D.A. Science and Technology Policy Institute estimates are almost 100% lower than those by the OECD, Space Foundation and the S.I.A. See Table 1.

Admittedly, there are considerable differences between the Science and Technology Policy Institute (STPI) estimates with those of the other three. Crane, Lynck, Lal and Wei [3] attribute the differences to a fair amount of double counting in the latter three estimates. In our view, the STPI estimates are more close to reality. According to the STPI, the space economy is divided into four segments:

- Government expenditures on Space
- Satellite services
- Space supplier's industry
- Space service user support industry

2.2. The U.K. Space Agency framework

On the contrary, the U.K. Space Agency (2020) [32] has developed another framework that decomposes the space economy into four segments: space manufacturing, space operations, space applications and ancillary services. See Table 2 for the details. Each of these segments has further sub-sectors.

The above framework has several advantages. These could be listed as follows:

i. It differentiates between stock and flow variables. All the components are flow variables that can be computed annually.
ii. It spells out the sub-components of all four segments. This helps us to get a comprehensive picture of a nation's space economy.
iii. In response to (i) and (ii), the possibility of double-counting, which is rampant in all the other frameworks, is virtually absent or minimal. This helps us to get a more realistic picture of the size and structure of the space economy.

2.3. US Bureau of Economic Analysis framework

The framework evolved by the Bureau of Economic Analysis of the U.S. Department of Commerce is the only official statistical agency in the world to attempt at producing a satellite account to the national account statistics of the country.[1] One of the satellite accounts is on the space economy. The agency has used a stakeholder approach in identifying the space component of all the sectors in the economy such as agriculture, utilities, construction, manufacturing (of which computer and electronic products and other transportation equipment), wholesale and retail trade, transportation and warehousing, Information (of which wired telecommunications carriers, satellite telecommunications), Finance, insurance, real estate and leasing, professional and business services, educational services, health care and social assistance, arts, entertainment, recreation, accommodation and food services, other services except government, the federal and state local budgets on Space (Highfill, Jouard and Franks, 2020 [8]). They have used the supply and use tables for arriving at national accounts characteristics of the space economy such as its gross output, value added, employment and compensation (or wages). Estimates of gross output and value added are available at both current and constant prices. The empirical aspects of this exercise are discussed in Section 5.1 where we discuss the size of India's space economy in a comparative fashion. The framework, while consistent with the national accounts methodology, is difficult to be operationalised as detailed supply and use tables are hard to come by for independent researchers.

Table 1
Different estimates of the size of the world space economy (Billions of USD).

Agency	2013	2016
IDA Science and Technology Policy Institute	155.70	166.30 (6.80)[a]
Satellite Industry Association (SIA)	320	339.10 (5.97)
Space Foundation	314.20	329.30 (4.80)
OECD	320	

Source: Crane, Lynck, Lal and Wei (2020), European Space Policy Institute (2020).
Note:
[a] Figures in parentheses indicate percentage change in 2016 compared to 2013.

In this paper, we propose to use an eclectic framework. For describing the space economy, we use the OECD framework, and for measuring its structure and size, we use the one used by the U.K. Space Agency. Such a mixture will enable us to map India's space economy comprehensively.

A recent phenomenon is the investments in the space economy by private sector enterprises.[2] In our present framework, this is captured in all the three sub sectors of the space economy: space manufacturing, operations and applications. The entry of offshore development centres and start-ups in the space sector has also happened recently. However, there are no official or private estimates of the sales of these two entities. To the extent that these are not included in our framework leads to a slight underestimation of the overall size of the space economy.

3. Engagement with the literature

The literature on the economic aspects of the space sector is small but growing. Most of the research on the space sector has been published in specialised journals on the aerospace industry like *Acta Astronautica*. However, issues related to the space industry have now been mainstreamed with Weinzierl (2018) [33] where the evolution of the US space sector over the years has been systematically documented: in the initial years, the US space sector was dominated by the state (read as by NASA and Department of Defence), but over time, the sector has been thrown open to the private sector. Nothing is more telling than the continuous reduction of the size of the U.S. space budget, which used to be in the region of 0.70% of her G.D.P. in the 1960s and in the 2000s, it is only about 0.20%. The Space vacated by the state has been taken over by the private sector, and there are now private sector enterprises having the full range of capability in all the three segments of the

[1] The System of National Accounts (SNA) provides the international standards for compiling macro-economic statistics. In addition to core set of accounts, the 2008 SNA introduced satellite accounts, which are linked to the Central framework of national accounts. The BEA has attempted satellite accounts in the economies of space, digital, ocean and outdoor recreation etc. See Highfill, Jouard and Franks (2020) [8].

[2] According to the present Foreign Direct Investment (FDI) policy, "FDI up to 100% is allowed in satellites-establishment and operation, subject to the sectoral guidelines of the Department of Space/ISRO, under the government route". However the Foreign Investment Facilitation Portal (https://fifp.gov.in/) where data on industry-wise FDI investments are reported does not contain any data on FDI in the space sector in India. Press reports indicate the likelihood of further relaxation of the FDI policy in the years to come.

Table 2
Structure of the space economy.

Space manufacturing
1. Launch, vehicles and subsystems
2. Satellites/payloads/spacecraft and subsystems
3. Scientific instruments
4. Ground segment systems and equipment
5. Suppliers of materials and components
6. Scientific and engineering support
7. Fundamental and applied research
8. Space test facilities
Space operations
1. Launch services
2. Launch brokerage services
3. Proprietary satellite operation (incl. sale/lease)
4. Third-party ground segment operation
5. Ground station networks
6. In-orbit servicing
7. Debris removal
8. Space surveillance and tracking (SST)
9. Space tourism
10. In-space manufacturing
11. Spaceports
Space applications
1. Direct-to-home (DTH) broadcasting
2. Fixed satellite communication services
3. Mobile satellite communication services
4. Location-based signal service providers
5. Supply of user devices and equipment
6. Processors of satellite data
7. Applications leveraging satellite signals/data
8. Other
9. Non-DTH Space Applications segment subtotal
Ancillary services
1. Launch and satellite insurance (including brokerage) services
2. Legal and financial services
3. Software and information technology (IT) services
4. Market research and consultancy services
5. Business incubation and development
6. Policymaking, regulation and oversight

Source: U K Space Agency (2020).

space economy such as space systems development, launch services and satellite operations. In fact, this move from the state to the market appears to have enlarged the size of the U.S. space economy, although the paper itself does not present any estimates of the size of the U.S. space economy.

In this section, we will undertake an issue-specific survey of the more scholarly work on the Indian space sector. This is because we have already reviewed the literature that focused on the measurement aspects of the space economy (see Section 2 above). Our primary purpose in this paper is to measure the size of India's space economy and to understand the larger economy that space technology would have created. The following six issues have come up for inquiry in the India-specific literature. The problems are not mutually exclusive, and the same author may have covered more than one issue. The issues and the authors who have dealt with them are:

- Historical evolution of space research in India (Raj, 2000; Suresh 2021) [28,30].
- Development of specific space technologies such as launch vehicles and liquid propulsion (Raj, 2000; Nagappa, 2016) [22,28].
- Role of the state in the creation and nurturing of space research and also commercialisation of space research (Baskaran, 2005, Prasad, 2021) [1,26].
- Analysis of the space budget (Mani, 2013; Kasturirangan and Shaijumon, 2021) [14,19].
- Cost-Benefit analysis of space research (Sankar, 2007; Murthy, Sankar and Madhusudhanan, 2007; Kasturirangan and Shaijumon, 2021) [14,21,29].
- Analysis of specific segments of the space economy-the case of remote sensing technology (Dayasindhu and Chandrasekhar, 2004, Jayaraman, Parihar, and Srivastava, 2008) [4,13].

A review of these studies reveals the following:

- None of the studies has addressed itself to measuring the size and structure of the space economy and that too using an internationally accepted framework for measurement.
- Except for a few, most studies are more than a decade old. Many changes have taken place in space technologies in India and abroad during the last decade. When these studies were conducted, satellite navigation was not available as an indigenous technology. Globally, navigation and positioning services provide the largest economic indirect impact among all space technologies. For instance, the private sector has played a much more significant role in the previous ten years in developing the technologies for manufacturing satellites, launching vehicles and launching satellites.
- Space technology has started to diffuse into many new domains such as location services and home entertainment, in addition to conventional domains such as communications, weather forecasting, disaster management, tele-education and telemedicine. In short, a much larger civilian market for space technologies has emerged.
- There has also been a gradual withdrawal of the state from the space economy globally, which is reflected in the decline in the growth rate of budgetary support in India. However, in the short-term, there is expected to be an increase to support *Gaganyaan*, the human spaceflight programme, announced for commemorating 75 years of Indian independence.

Considering the above, there is a severe gap in our understanding of the contours of our space economy and its size in some precise terms. Such an understanding will go a long way towards designing policies for the space sector in India that are more evidence-based. This is the gap that our present paper is attempting to fill.

4. Contours of India's space economy

In this section, we estimate the size and structure of the space economy of India by employing an eclectic framework; we first describe the space economy in terms of the OECD framework.

4.1. Inputs

4.1.1. Government budget

The intensity of space funding or the ratio of space budgets to GDP is one of the most widely used indicators describing the space economy (OECD, 2019) [25]. See Table 3 for India's space budget intensity computed both in nominal and real terms. See Table 3.

Right through its inception in 1968, Indian Space Research Organisation (ISRO) has been funded by the central government through generous budgetary support. It is one of the country's top government grants receiving research councils. However, from around 2011–12, government budgets have been reduced as a percentage of GDP to about 0.05% and have remained at that level since then.

Table 3
India's space budget intensity, 2000–01 through 2020–21.

	Nominal budget (Rs in Crores)	The nominal rate of growth (%)	Real budget (Rs in crores)	The real rate of growth (%)	**Intensity of the space budget** in nominal terms (%)	Space budget intensity in real terms (%)
2000–01	1905.4		2239.94		0.09	0.09
2001–02	1914.15	0.46	2180.12	−2.67	0.08	0.08
2002–03	2165.43	13.13	2377.96	9.07	0.09	0.09
2003–04	2275.7	5.09	2405.99	1.18	0.08	0.08
2004–05	2540.78	11.65	2540.78	5.60	0.08	0.08
2005–06	2667.6	4.99	2559.17	0.72	0.07	0.07
2006–07	2988.67	12.04	2694.15	5.27	0.07	0.07
2007–08	3278	9.68	2794.14	3.71	0.07	0.07
2008–09	3493.57	6.58	2740.44	−1.92	0.06	0.06
2009–10	4162.96	19.16	3078.83	12.35	0.06	0.06
2010–11	4482.23	7.67	3041.69	−1.21	0.06	0.06
2011–12	3790.78	−15.43	3790.78	24.63	0.04	0.04
2012–13	4856.28	28.11	4499.29	18.69	0.05	0.05
2013–14	5168.95	6.44	4509.96	0.24	0.05	0.05
2014–15	5818.37	12.56	4912.91	8.93	0.05	0.05
2015–16	6920.01	18.93	5712.88	16.28	0.05	0.05
2016–17	8040	16.18	6441.57	12.76	0.05	0.05
2017–18	9130.57	13.56	7020.62	8.99	0.05	0.05
2018–19	11192.66	22.58	8261.68	17.68	0.06	0.06
2019–20	13033.29	16.44	9335.64	13.00	0.06	0.06
2020–21	9500.00	−27.11	6501.22	−30.36	0.05	0.05

Source: Union Budget and Economic Survey (various issues), Government of India.
Note: Real values are in 2004–2005 prices up to 2010–11 and at 2011–12 prices from 2011–12 until 2020–21; Nominal values have been converted to real values using the implicit G.D.P. deflator.

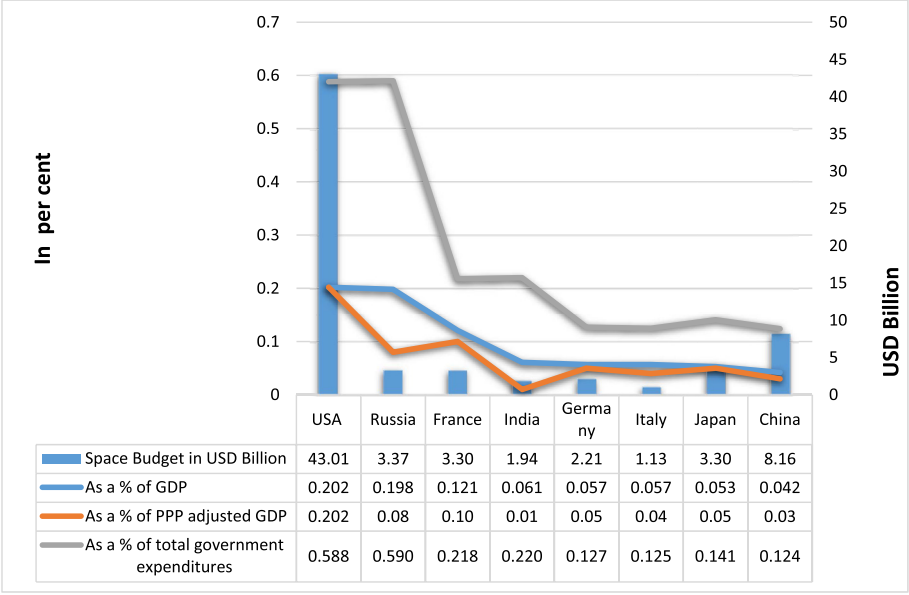


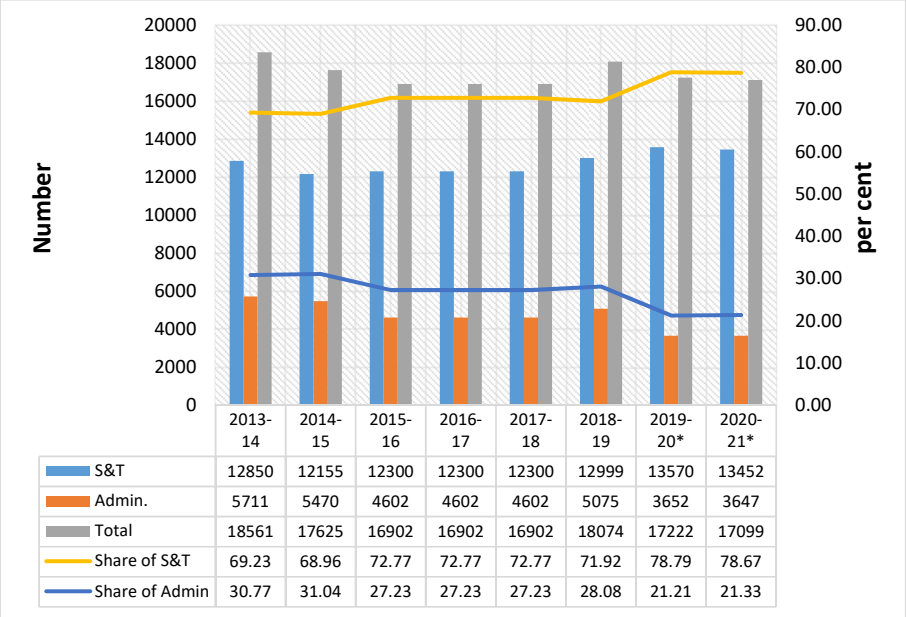
Fig. 1. Space budget across major space-faring nations, 2019. Source: European Space Policy Institute (2020).

The budget growth rate has fluctuated throughout the period under consideration. In nominal and real terms, the budget has turned negative in 2020–21 due to the difficult economic conditions caused by the pandemic. In short, although space research has garnered support from the government, this financial support has not been registering secular growth. But India's space budget is one of the highest in the world and especially among the major space-faring nations (Fig. 1).

4.1.2. Human capital

One of the most critical assets of the space sector is its well-trained human resource. The space sector in India, represented by the ISRO, is the main employer and has 18 centres and units distributed across more than 44 locations in the country. Fig. 2 presents the trends in the total human resources employed by ISRO, giving the total breakdown into the S&T and administration-based workforce. Over time, the total human resource employed in ISRO has been reduced. But the reduction has been more at the administration level, so much to say that ten percentage points have reduced its share in the total, and the share thus vacated has been taken over by the S&T staff. Thus, ISRO appears to have bucked the general trend in India's scientific establishment, where several scientific organisations have undergone downsizing by reducing their S&T staff.

In contrast, those in administration have managed to retain their positions. ISRO has thus been successful in weathering the general move in the country towards downsizing in public sector



	2013-14	2014-15	2015-16	2016-17	2017-18	2018-19	2019-20*	2020-21*
S&T	12850	12155	12300	12300	12300	12999	13570	13452
Admin.	5711	5470	4602	4602	4602	5075	3652	3647
Total	18561	17625	16902	16902	16902	18074	17222	17099
Share of S&T	69.23	68.96	72.77	72.77	72.77	71.92	78.79	78.67
Share of Admin	30.77	31.04	27.23	27.23	27.23	28.08	21.21	21.33

Fig. 2. Human capital in India's space economy. Source: Indian Space Research Organisation (various issues).

employment. Furthermore, to offset the fall in the availability of good quality S&T personnel trained in aerospace engineering, ISRO established the Indian Institute of Space Science and Technology (IIST) in 2007. Not considered here is the substantial contractual manpower deployed by ISRO for technical and non-technical activities.

4.1.3. *Assets of the space sector*

This consists of tangible and intangible assets. Tangible assets are satellites, satellite launch vehicles, capital goods employed to manufacture or assemble launch pads, tracking stations, land and buildings. Intangible assets include computer software, patents, copyrights, motion picture films, customer lists, import quotas, customer or supplier relationships, customer loyalty, market share and marketing rights. As far as the space sector is concerned, data on both tangible and intangible assets are very hard to come by. Of the two, tangible assets are relatively easier to estimate. These are crude estimates as they have been arrived at using several simplifying assumptions and consider only satellites and satellite launch vehicles.

The assets have been growing on an average of around 19.5% since 2000. The rate has decelerated significantly over the last two years. We have also measured the assets of the space sector as a percentage of (a) the net capital stock of the public sector communications and broadcasting services industry; (b) the net capital stock of the public sector industry as a whole. These (a) and (b) are calculated from the Central Statistical Organisation (2021) [20] and are presented in Fig. 3. The assets of the space sector have increased from about 6% in 2012 (of net capital stock of the public sector communications and broadcasting services industry) to almost 14% in 2020, while its share as a per cent of the whole public sector industry stock has been more or less constant at about 0.17% (Fig. 3).

As noted above, there is some underestimation of tangible assets. Given that ISRO has developed all its technologies strictly through in-house research and development (R&D) efforts, the space sector in India would have accumulated a considerable amount of intangible assets in the form of intellectual properties of various kinds. However, given the strategic importance of space research and its dual-use, much of it may have remained in a tacit rather than codified form. The estimation of intangible assets in space research is an important exercise that future research can consider.

4.2. *Activities and outputs*

4.2.1. *Space manufacturing industry*

The country has a thriving space manufacturing industry initially mainly confined to the public sector. The country manufactures its satellites and satellite launch vehicles, and many components are used to manufacture them. The private sector was initially confined only to the manufacture of parts and materials. Still, recently, policies have opened up the private sector participation in manufacturing satellites and launch vehicles. With the incentivisation of the start-up ecosystem, several start-ups have sprung up in the space manufacturing and services arena. The three vertical policies that have a direct bearing on the growth of space manufacturing have been (i) the Satellite Communication Policy of 1997; (ii) the Remote sensing policy of 2011; (iii) the Draft Spacecom Policy of 2020. See Box 1 for the details of these policies.

Space product manufacturing consists of satellites and launchers of 7 different types. See Table 4. The most important ones are communication and earth observation satellites and Polar Satellite Launch Vehicle (PSLV) and Geosynchronous Satellite Launch Vehicle (GSLV) series of launch vehicles.

Based on the number of satellites and launch vehicles fabricated and launched, we have estimated the value of satellite and launch vehicle manufacturing to arrive at the total value of space manufacturing. See Table 5.

Since 2017, there has been a significant deceleration in the manufacturing of space products. There are four distinct features of the manufacturing of the space products: (i) the hardware is manufactured domestically with main components also sourced domestically; (ii) most of the hardware is fabricated by ISRO itself in

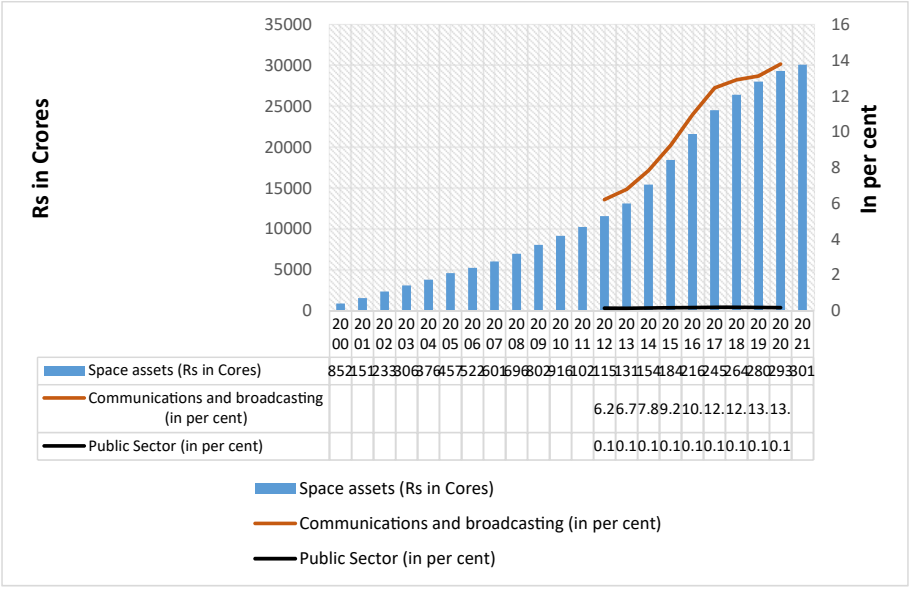


Fig. 3. Estimated tangible assets of India's space economy. Source: ISRO data & Central Statistical Organisation (2021).

Box 1
Sector-specific vertical policies having an impact on space manufacturing and services in India

Satellite communication policy, 1997

- The government approved a policy framework for satellite communication in India in 1997. The government approved the norms, guidelines and procedures for implementing the Policy Framework for Satellite Communications in India in 2000. The policy aims to build capability and develop the country in satellites. To fulfil the purpose, the INSAT program plays an important role. It is managed by the INSAT Coordination Committee (I.C.C.) with technical support from its Technical Advisory Group (T.A.G.).[8]

Remote sensing policy, 2011 and 2020

- In 2011, India adopted the Remote Sensing Data (R.S.D.) Policy. The nodal agency for all actions under the policy is the Department of Space (DoS) of the Government of India. The National Remote Sensing Centre (NRSC) of ISRO/DOS is tasked with obtaining and circulating all satellite remote sensing data in India, both from Indian and foreign satellites. Antrix Corporation Ltd. (of D.O.S.) will be responsible for the licence grant for the acquisition/distribution of I.R.S. data outside India.[8]

Draft Spacecom policy, 2020

- Department of Space published "Draft- Spacecom Policy-2020". The policy aims to meet the growing demand of the nation's space-based communication requirements. This will boost the government's initiatives towards Self Reliant India (*Aatmanirbhar Bharat*) that will drive the focus on "ease of doing business" and encourage healthy competitiveness in the growth of the national economy

Source: Government of India, https://www.makeinindia.com/sector/space (accessed on January 20, 2022).

its in-house facilities; (iii) the entire disembodied technology for its design, manufacturing and launching by ISRO itself through its in-house R&D efforts; (iv) space manufacturing is yet another instance of frugal innovations and manufacturing. The average cost of a satellite in India is about Rs. 330 Crores, and of launch vehicles is about Rs. 114 Crores.[3] These rates are only a fraction of the international cost of satellites and launch vehicles. This provides immense scope for India to emerge as a hub for manufacturing space products.

[3] The range is Rs 2 cores to Rs 1103 crores in the case of satellites and Rs 60 crores to Rs 367 crores in the case of satellite launch vehicles (Based on ISRO data).

Table 4
Types of spacecraft and launchers domestically manufactured and launched (Cumulative 1975–2021).

Satellites		Launch vehicles	
Type	Number	Type	Number
Communication	43	Sounding rockets	100
Earth observation	40	SLV	4
Experimental	8	ASLV	4
Satellite navigation	9	PSLV	53
Small satellites	2	GSLV	14
Scientific and exploration spacecraft	7	GSLV Mk III	4
University/Academic institute	12	RLV-TD	1
		Scramjet Engine-TD	1
Total	**121**		**81**

Source: Compiled from the website of the Indian Space Research Organization, https://www.isro.gov.in/spacecraft; and https://www.isro.gov.in/launchers (accessed on January 21, 2022).

4.2.2. Space telecommunications industry

As noted earlier, the space telecommunications sector is the most important commercial market for space technology. In India, space technology is relied on only for D.T.H. television and local services. India's basic mobile telephony and broadband distribution is based on terrestrial networks.

With the issuance of the 'Guidelines for obtaining the licence for providing D.T.H. broadcasting service in India', D.T.H. services were allowed in the country in 2001. These guidelines, among other things, prescribe the eligibility criteria, the procedure for obtaining the licence to set up and operate D.T.H. services in the country, and the basic terms and conditions/obligations reposed in the operators. One of the most important requirements for providing D.T.H. services in the country was that all D.T.H. service providers had to

Table 5
Trends in space manufacturing.

Year	Satellites (Rs. Crores)	Launch vehicles (Rs. Crore)	Total space manufacturing	Rate of growth (%)
1997	66		66	
1998	129.45		129.45	96.14
1999	233.84	56	289.84	123.90
2000	482.24	80	562.24	93.98
2001	535.01	132	667.01	18.63
2002	655.907	156	811.907	21.72
2003	602.147	136	738.147	−9.08
2004	572.137	128	700.137	−5.15
2005	633.337	176	809.337	15.60
2006	455.626	192	647.626	−19.98
2007	617.626	168	785.626	21.31
2008	686.913	262	948.913	20.78
2009	734.123	334	1068.123	12.56
2010	877.446	255	1132.446	6.02
2011	857.531	217	1074.867	−5.08
2012	992.337	331	1322.873	23.07
2013	1118.029	427	1545.297	16.81
2014	1636.089	677	2313.089	49.69
2015	2030.494	993	3023.494	30.71
2016	1937.244	1224	3161.044	4.55
2017	1701.22	1217	2918.42	−7.68
2018	1076.85	827	1903.65	−34.77
2019	1930.85	659.95	2590.8	36.1
2020	2038.31	1395.02	3433.33	32.52

Source: Estimates based on ISRO data.

Table 6
Trends in the number of D.T.H. Subscribers (in million numbers).

	Cable T.V. subscribers	D.T.H. subscribers	Total subscribers	Share of D.T.H. (%)	Rate of growth of D.T.H. (%)
2012	88	28.09	116.09	24	
2013	93	32.69	125.69	26	16
2014	99	37.19	136.19	27	14
2015	101	41.15	142.15	29	11
2016	102	58.53	160.53	36	42
2017	92	63.61	155.61	41	9
2018	98.5	67.53	166.03	41	6
2019	103	72.44	175.44	41	7
2020	103	70.26	173.26	41	−3

Source: Telecom Regulatory Authority of India.

use satellites commissioned by ISRO compulsorily.[4] Actual D.T.H. operations commenced in India in 2003. India has one of the most significant numbers of D.T.H. subscribers in the world. The total number of D.T.H. subscribers has now grown to around 70 million in 2020. However, the pandemic and the decrease in economic activity appear to have reduced the number of subscribers in that year (Table 6).

Interestingly, 41% of cable television subscribers are D.T.H. users. The D.T.H. services industry in India is very highly concentrated, with just four service providers. The degree of concentration as measured by the Herfindahl Index works out on average during 2016–17 through 2019–20 to be 0.28. However, with the arrival of online streaming in more recent times and especially during the pandemic years of 2020–21, the future growth of D.T.H may be adversely affected.

We have estimated the size of the space telecom sector by taking the sum of the sales revenue of mobile location services and that of D.T.H. While the actual D.T.H. sales revenues each year can be directly taken from the reported sates revenue of each of the four D.T.H. service providers, we have estimated those for the

Table 7
Trends in space telecom revenues (Values are in Rs. Crores at Current prices).

	Telecom	D.T.H.	Total space telecom	Gross output of communication and broadcasting services	Share (%)
2011–12	8112.7	7435.4	15548.1	351495	4.42
2012–13	8793.8	8364.26	17158.0	394042	4.35
2013–14	9266.1	9424.2	18690.3	450379	4.15
2014–15	10399.4	10735	21134.4	525349	4.02
2015–16	10627.9	12249.2	22877.1	548167	4.17
2016–17	10505.0	11706	22211.5	580089	3.83
2017–18	9186.6	15459	24645.7	631625	3.90
2018–19	11654.1	18017	29671.4	677310	4.38
2019–20	11967.0	12656	24622.6	706786	3.48
2020–21	13476.1	12414	25890.3		

Source: Telecom Regulatory Authority of India, Company Annual Reports, Central Statistical Organization (2021).

location services. We do it by taking 5% of the sales revenue of mobile telephony services as reported by the Telecom Regulatory Authority of India [31]. We measure the relative size of the communications and broadcasting services of the country by taking the percentage share of space telecom services in the former, See Table 7. The share of D.T.H. in the total space economy of India is about a third (details to be seen in Section 5), and it appears to be the single largest segment of the space economy mirroring the international pattern.

4.2.3. *Satellite earth observation sector*

With around 40 earth observation satellites, India is one of the leading countries in Asia with considerable technological capability in remote sensing. The 'Remote Sensing Data Policy of India, 2001 made the National Remote Sensing Centre (NRSC)[5] [43] with the

[4] They are also allowed to use the leased in capacity from foreign satellites by ISRO if sufficient capacity for their use is not available on ISRO satellites.

[5] Until 2008, the National Remote Sensing Agency was an autonomous organisation under the Department of Space. It was then converted into a government entity as one of the 41 centres of ISRO, due essentially to constraints faced by NRSC in dealing with programmes of national importance on account of its status of being a society.

sole authority to acquire and disseminate all remote sensing data in the country subject to specific guidelines stipulated to protect national security interests. The sale of data products by the NRSC has been very small (in relation to the capacity created) according to a report published by the Comptroller and Auditor General of India in 2011. According to this report, the sale of data products was Rs. 20.97 crores in 2003–04, and it increased by Rs. 42.63 crores in 2007–08 (Comptroller and Auditor General, 2011) [2]. No specific data on remote sensing data products have been available since 2008 as NRSC's sales revenues are not reported separately as it is now a part of ISRO. The policy also granted the right to disseminate data acquired by Indian satellites outside India to Antrix. While onboard recorders can collect and store data over any part of the globe, large data acquisition requires local ground stations. Antrix had licenced many such ground stations globally, starting for IRS-1C. Of late, the adoption of open R.S. data policy by the U.S.A. (USGS for Landsat) and E.U. (for Sentinel constellation) has led to a drop in demand, and there has been a rise of commercial remote sensing companies providing very high resolution data (<1 m resolution), which is not offered by ISRO to outside world. According to many analysts, the primary constraint to large scale usage of remote sensing data was discouraged by specific provisions in the Remote Sensing Data Policy of 2001. Specifically, for high-resolution satellite images, permissions were required from a High-Resolution Image Committee. The new Remote Sensing Data policy of 2011 seems to have remedied this problem: the threshold resolution of 5.8 m and above was now reduced to 1 m and above. This relaxation is sure to have increased the size of the earth observation market in India, although we do not have precise numbers of its size. However, a significant change in remote sensing that would essentially increase the size of the market for location-based data products occurred with the announcement of a slew of vertical policies, which may change the complexion of the industry. The first and foremost in this series of policies (See Table 8 for a mapping of these policies) is the 2020 space-based remote sensing policy of India, which is likely to enhance the market for space-based remote sensing data (Department of Space, 2020) [6]:

- Promote Indian Industries to carry out space-based remote sensing activities within and outside India.
- Enable easy access to space-based remote sensing data except for sensitive data and information.
- Focus on realising space-based remote sensing systems to meet the needs of the country, which cannot be met effectively, affordably and reliably by commercial entities, either due to national security concerns or economic factors.
- Provide a timely and responsive regulatory environment for the commercial Indian industry to establish and operate space-based remote sensing systems.

There are two main differences between this draft policy and the earlier ones. First of all, the new policy is exclusively concerned with space-based remote sensing while the earlier ones are concerned itself to addition to aerial remote sensing using aircraft; (ii) the new policy aims to develop a larger market for space-based remote sensing products explicitly by involving what it refers to as non-government private entities to design, launch and maintain remote sensing satellites and also allow them to establish Telemetry, Tracking & Command and Satellite data reception stations both with and outside India. In addition, they can offer the capacity for commercial and societal applications within India and abroad. In other words, the draft policy seeks to end ISROs monopolistic position in the entire supply chain in space-based remote sensing, except for R&D in developing new technologies for remote sensing.

Given the progress in implementing these public policies, the satellite earth observation sector is poised to increase the size of India's space economy. The Indian geospatial industry, which is a service industry and is substantially fuelled by high-resolution satellite data, is also estimated to be worth $ 4 billion.

4.2.4. *International trade in space products from India*

India has followed a policy of self-reliance and import substitution as far as its space sector is concerned. In fact, its space sector is an important example where the country has succeeded in developing its technological capability in a wide variety of space technologies. Space products are essentially satellites and satellite launch vehicles, and the components employed in the manufacture of these two products. While satellites and launch vehicles are manufactured within the country, some components, especially proprietary electronic components, are imported from abroad. India's international trade in space products (H.S. Code: 880260) is presented in Table 9. Except for 2008, the country has had a negative trade balance over the years.

Both imports and exports do not show any trend but only violent year to year fluctuations.

4.2.5. *Innovations in the space industry in India*

We employ three indicators to measure innovations in the space industry. These are (i) shares of scientific publications in space literature based on SCOPUS custom data from Elsevier and compiled by OECD (2019)[6] [25]. This refers to only one year, namely 2016 but across all the major spacefaring nations (Table 10); (ii) number of patents granted to ISRO (Fig. 4); (iii) number of new space startups in India (Table 11).

4.2.5.1. Scientific publications. The first indicator of knowledge production is the number of scientific publications in space literature indexed by well-known bibliographic databases such as SCOPUS, see Table 10. India has a 2% share of the scientific literature in space. As this output would include publications from research and higher education institutions in addition to ISRO, funding support from scientific and higher education funding agencies has not been considered for this component.

4.2.5.2. Patents. Patents may not be the right indicator to measure new knowledge production in India's space economy. Given the dual-use nature of this technology, some or a large proportion of the new knowledge generated may be kept as a trade secret and therefore not codified. However, with the increasing privatisation of space activities in India, ISRO may increasingly seek intellectual protection in patents. We discuss both the stock of patents with ISRO and also its flow during more recent times. However, we do not have data on the implementation of these patents.

4.2.5.2.1. Stock of patents. According to ISRO, as of 2022, its I.P.R. portfolio consists of more than 270 patents, 45 copyrights, and ten trademarks. Of these three I.P.R.s, the stock of patents has been decomposed into its various components technology-wise (Fig. 4). The patents have been grouped under launch vehicles, satellite communication, computer network and earth observation. The lion's share of the patents is in launch vehicles and chemical compounds, polymers and electromechanical structures. This is followed by satellite communication and within it antenna systems. The least number of patents is in earth observation, which

[6] According to a question answered in the Lok Sabha (the lower house of the Indian parliament) Indian scientists have published a total of 540 articles on space in refereed journals using the data from Hubble Telescope. See Lok Sabha (2022).

Table 8
Basket of new public policies promoting the diffusion of remote sensing data, 2020–2021.

Name of policy	Year	Main features
(Draft) Space-based remote sensing policy of India[a]	2020	• The policy recognises the increasing demand for satellite data to meet the country's requirements for developmental and commercial activities • Allow private sector entities to provide end-to-end space-based remote sensing service domestically and abroad on a commercial basis • It authorises the private sector entities to share commercial data with resolutions better than 5 m and up to 50 m freely to any users within India • The Department of Space, through ISRO, will continue to emphasise R&D in space-based remote sensing • The Indian National Space Promotion and Authorisation Centre (IN-SPACe) will be a single-window regulatory agency to provide authorisations and permissions for all Space-based remote sensing activities in India
Guidelines for acquiring and producing geospatial data and geospatial data services including maps[b]	2021	• It liberalises the pre-existing regulations around geospatial data in India • Paves the way for the unrestricted environment for its usage across diverse sectors of the economy • Encourages the development of geospatial entrepreneurship in the country • Facilitates the diffusion of geospatial technologies in the country.
(Draft) National Geospatial Policy[b]	2021	• "A plan will be formulated for building high-resolution satellite constellations to provide data with a high revisit, establishing ground infrastructure for data acquisition, processing, generation of data products and reliable dissemination mechanism, for constellation of microwave Synthetic Aperture Radar (S.A.R.), imaging scatterometers, imaging microwave radiometers, altimeters, high resolution panchromatic, multispectral and hyperspectral remote sensing data with quick turn-around-time. Production and dissemination of such data sets will be harmonised with the National Remote Sensing Data Policy."
(Draft) Indian Satellite Navigation Policy[a]	2021	• It incentivises both NavIC and GAGAN • NavIC (Navigation with Indian Constellation) is the operational name of the Indian Regional Navigation Satellite System (IRNSS) developed by ISRO. Unlike G.P.S., a Global tracking constellation, Navic has focused primarily on India and adjoining regions (1500 km around India). • GAGAN—it is a system to improve the accuracy of a global navigation satellite system (GNSS) receiver by providing reference signals
The Drone Rules[c]	2021	• This liberalises domestic manufacturing and the use of drones that can lead to the increased use of this technology for mapping and surveying.

Source: Compiled from
[a] ISRO https://www.isro.gov.in/sites/default/files/spacers_policy_ngp_2020_draft.pdf, and https://www.isro.gov.in/sites/default/files/satnav_policy-29.pdf).
[b] Department of Science and Technology (https://dst.gov.in/sites/default/files/Final%20Approved%20Guidelines%20on%20Geospatial%20Data.pdf; and https://dst.gov.in/sites/default/files/Draft%20NGP%2C%202021.pdf).
[c] Ministry of Civil Aviation (https://www.civilaviation.gov.in/en/ministry-documents/rules).

Table 9
International trade in space products from and to India (Millions of USD).

	Imports	Exports
2007	78.56	0
2008	8.91	32.27
2009	0.91	0.65
2010	2.25	
2011	18.82	0.02
2012	194.19	0.96
2013	30.97	36.65
2014	153.02	169.77
2015	138.35	0
2016	72.55	0
2017	40.86	0
2018	71.50	1.08
2019	26.57	0
2020	39.37	0.03

Source: Compiled from U.N. Comtrade.

Table 10
Relative share of India in scientific publications in space research, 2016. (Based on SCOPUS-indexed publications).

	Share (%)
USA	21.5
China	8.8
Germany	8.3
UK	7.6
France	5.9
Italy	4.8
Canada	2.9
Netherlands	2.9
Australia	2.77
India	2
Korea	1.4
Mexico	1.3

Source: OECD (2019).

worked as an independent unit 2008. It is not immediately apparent to which jurisdiction these patents were filed. A patent search at the USPTO showed that ISRO had only a cumulative total (1979 through 2020) of just nine patents there.[7] Of these 9 patents, 5 patents have been granted after 2015. Regarding the quality of these patents (as judged by their citation), only one has received some 17 citations.[8] It is conceivable since ISRO was developing technologies for its captive use in space systems, filing patents did not get priority, and second, given barriers to technology sharing, indigenisation would lead to the creation of local capability, which would not be patentable.

4.2.5.2.2. Patent flows. Data on patents granted to ISRO are available only from 2013–14 to 2019–20 (Fig. 5). It has been showing a rising trend in more recent times.

[7] Patentview.org, https://datatool.patentsview.org/#detail/assignee/5c140a20-83e2-48e8-952e-1f0fdb202aa5 (accessed on January 25, 2022).

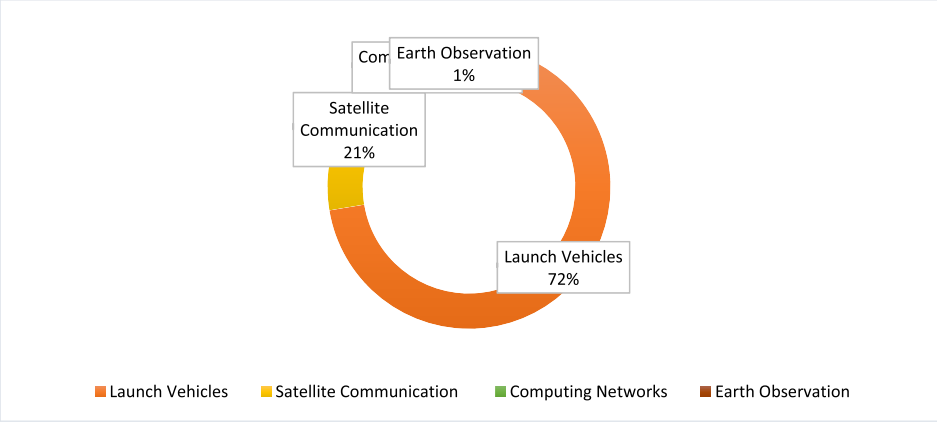


Fig. 4. Technology-wise stock of patents with ISRO, cumulative 1973 through 2020. ISRO, Indian Space Research Organisation. Source: Insights, https://insights.greyb.com/isro-patent-portfolio/ (accessed on January 25, 2022).

Table 11
Number of startups in India's space economy (2007–2021).

Name of the startup	Tier	Location
1. Skyroot Aerospace	Launch Vehicle	Hyderabad
2. Agnikul cosmos	Launch Vehicle	Chennai
3. Dhruva space	Satellite/spacecraft sub systems	Hyderabad
4. Xovian	Satellite/spacecraft sub systems	Bangalore
5. Astrome Technologies	Satellite/spacecraft sub systems	Bangalore
6. Bellatrix Aerospace	Satellite/spacecraft sub systems	Bangalore
7. Satellize	Satellite/spacecraft sub systems	Mumbai
8. Astrogate labs	Satellite/spacecraft subsystems	Bangalore
9. Manastu space	Satellite/spacecraft sub systems	Mumbai
10. Pixxel	Satellite/spacecraft sub systems	Bangalore
11. Garuda Aerospace	Satellite/spacecraft sub systems	New Delhi
12. D'Vine research Lab	Satellite/spacecraft subsystems	Bangalore
13. Saankhya Labs	Satellite applications	Bangalore
14. Skylo Technologies	Satellite applications	Bangalore
15. Earth2Orbit	Satellite applications	Bangalore
16. Aniara	Satellite applications	Bangalore
17. Satsure	Satellite applications	Switzerland and Bangalore
18. Numer8	Satellite applications	Mumbai
19. Blue Sky Analytics	Satellite applications	Haryana
20. Kawa Space	Satellite applications	Mumbai
21. Mist E.O.	Satellite applications	Thiruvananthapuram
22. Team Indus (Axiom Research Lab)	Other	Bangalore
23. Nopo Nano Technologies	Other	Bangalore
24. Valles Marineris International	Other	Chennai
25. Rocketeers	Education	Bangalore
26. Society for Space Education Research and Development (SSERD)	Education	Bangalore

Source: Based on Prasad (2021).

Detailed data on the licencing of these patents are not available. However, ISRO reports indicate the transfer of around 300 technologies to domestic firms; some of the start-ups are based on these ISRO developed and commercialised technologies. ISRO has, over the years, formalised its technology processes and, in this context, has published a revised set of guidelines for technology transfer in 2020.[9] The 2020 guidelines have identified three different channels of technology transfer in space technologies. First is the conventional licencing of technology transfer channel wherein ISRO develops space technologies through its in-house R&D efforts and then transfers it to enterprises in India's industrial establishment. ISRO has developed eight space technologies through in-house R&D efforts (Box 2). But there is no further data on whether these technologies have been licenced and productionised.

The second is joint technology development projects with the industry. The third is the sponsored R&D projects with reputed academic institutions such as the I.I.T.s and IISc. This is known as the RESPOND programme. Technologies developed through these three channels will be transferred through the newly established technology transfer organisation of the Department of Space known as the New Space India Limited. Given the more explicit articulation contained in the 2020 guidelines, one would expect a much higher incidence of technology transfer from ISRO to domestic industry in the years to come.

4.2.5.3. New start-up formation. Maturing of space technologies and the widespread diffusion of Information and Communication technologies have lowered barriers to entering the space industry. This, combined with the incentivisation of the country's start-up ecosystem, has led to the emergence of several space-technology based new firms. There are no official estimates of the number of such start-ups. However, some data on the number and type of space start-ups are available from Prasad (2021)[10] [26], see Table 11. Listed here are 26 space start-ups in 4 different sub-sectors, namely launch vehicles, satellite subsystems, satellite applications, etc. Most of the firms are satellite subsystems, and the major location of a space cluster is in the city of Bangalore, where ISRO has its headquarters and is among the top ranking locations for I.T. start-ups. Although most of these startups have been initiated by young engineers, quite a few have retired ISRO employees as senior advisers and technology officers.

With more formal technology transfer envisaged through New Space India Limited, one could expect more such startups to be created in the future.

[9] See ISRO (2020).

[10] According to Lok Sabha (2022a), there are 75 start-ups listed under space technology in the website of Start Up India.

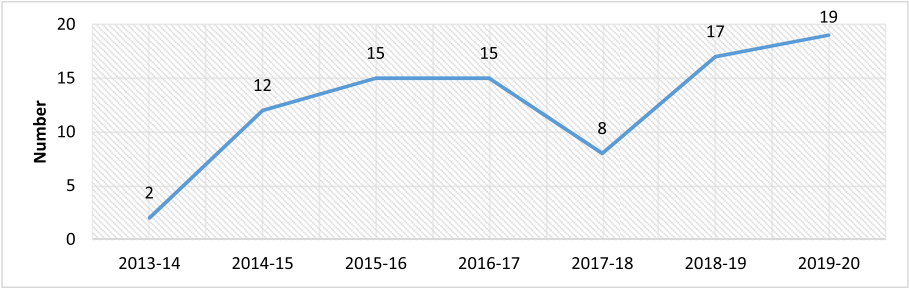


Fig. 5. Patents granted to ISRO. ISRO, Indian Space Research Organisation. Source: Lok Sabha (2016) and Lok Sabha (2020).

Box 2
Technologies developed by ISRO

- Space Grade-Li-ion cells
- NavIC message receiver
- Mini synthetic aperture radar
- Thermal sensors
- Two-channel digital mono pulse tracing receiver
- Optimal imaging system
- Pedcot liner
- Two-way M.S.S. terminal

Source: Lok Sabha (2020).

4.2.6. Insurance market for space

The following three propositions determine the size of the insurance market for space in India. First, all Indian satellites launched from India by ISRO are not insured as the satellites belong to the Government of India. Second, Indian satellites (e.g. the Indian National Satellite System (INSAT) and Geosynchronous Satellite System (GSAT) series of satellites) launched from abroad are insured. Third, foreign satellites launched by Antrix/ISRO on commercial terms are insured by satellite owners, and most likely, the insurance companies, in this case, are based abroad. So, the size of India's insurance market for space products is almost primarily determined by the second proposition. We have compiled the premium paid for most communication and one earth observation satellite launched from abroad from 1990 to 2020, and the cumulative total of the premium paid is Rs. 1463 crores. The average premium paid per satellite is about Rs. 63 crores. See Table 12.

Table 12
An estimate of the size of India's market for space insurance (Rs in Crores).

Name of the Satellite	Launch date	Satellite cost	Launch cost	Insurance premium
INSAT-1D	Jun 12, 1990	80	117	18
INSAT-2A	Jul 10, 1992	143	157	26
INSAT-2B	July 23, 1993	171	157	31
INSAT-2C	Dec 7, 1995	210	157	32
INSAT-2D	Jun 4, 1997	225	157	36
INSAT-2E	Oct 02, 1999	293	157	43
INSAT-3B	Mar 22, 2000	150	275	70
INSAT-3C	Jan 24, 2002	150	275	70
INSAT-3A	Apr 10, 2003	150	275	70
INSAT-3E	Sep 28, 2003	150	275	70
INSAT-4A	Dec 22, 2005	300	230	92
INSAT-4B	Mar 12, 2007	210	230	60
GSAT-8	May 21, 2011	259	297	30
GSAT-10	Sep 29, 2012	344	356	50
GSAT-7	Aug 30, 2013	270	420	65
GSAT-16	Dec 07, 2014	270	491	90
GSAT-15	Nov 11, 2015	210	510	90
GSAT-18	Oct 06, 2016	210	550	90
GSAT-17	Jun 29, 2017	1013	550	90
GSAT-11	Dec 05, 2018	578.53	721	90
GSAT-31	Feb 06, 2019	256	800	90
GSAT-30	Jan 17, 2020	256	800	90
INSAT-3D	July 26, 2013.	351	490	70
Total				1463

Source: Own compilation based on answers to questions in the Lok Sabha and Rajya Sabha.

5. Size and structure of India's space economy

5.1. Size

Employing a judicious mixture of the two frameworks, OECD and the U.K. Space Agency, we have estimated the size of India's space economy and expressed it as a percentage of the country's G.D.P. We have estimated both high and low estimates of the size of the space economy, which we denote as unadjusted and adjusted size estimates, respectively, see Table 13.

We have confined our size estimation to the ten years 2011–12 through 2020–21 due to the availability of the reasonably good quality of data on the three segments, namely space manufacturing, operations and applications, which together make up the space economy. The size has increased with fluctuations from year to year until 2018–19. After that, it has been declining essentially due to reductions in the space budget intensity. The size is currently about USD 5 billion (unadjusted) and about USD 4 billion (adjusted) in U.S. dollar terms. The size of the space economy about the size of the overall economy is about 0.23% (unadjusted) or 0.19% (adjusted). The relative size also has declined over the last two years.

How does our estimate of the size of India's space economy compare with other estimates available? The only different estimate of India's space economy that we are aware of is by the accounting and consultancy firm, PwC (2020) [27]. The PwC study has estimated it to be USD 7 billion in 2019. ISRO has used these exact estimates in its booklet, *Unlocking the space sector* (see the study by ISRO, Unlocking the Space Sector, on the Path to Atmanirbhar Bharat [12]); the *Make in India* programme projects the space economy to reach a size of USD 50 billion by 2024 [34]. At these estimates, according to ISRO (2021) [12], India's space economy is about 2% of the global space economy. Although PwC does not specify its methodology, our estimates of the space economy for 2019 are more or less similar to those by the PwC: ours is 6.21 while that of the PwC is USD 7 billion. However, it has since declined during 2019–20 and 2020–21 to about USD 5 billion. At this level, India's space economy of USD 5 billion in 2021

Table 13
Estimated size of India's space economy.

Year	Size in Rs. Crores-unadjusted	Size in Rs. Crore-adjusted	Size in Million USD- unadjusted	Size in Million USD- adjusted	Size as a per cent of G.D.P.-unadjusted	Size as a per cent of G.D.P.- adjusted
2011–12	22624	19012	4721	3967	0.26	0.22
2012–13	24995 (10)	21177 (11)	4594 (−3)	3892 (−2)	0.25	0.21
2013–14	27497 (10)	23535 (11)	4545 (−1)	3890 (0)	0.24	0.21
2014–15	32471 (18)	27914 (19)	5311 (17)	4565 (17)	0.26	0.22
2015–16	35133 (8)	29516 (6)	5366 (1)	4508 (−1)	0.26	0.21
2016–17	35241 (0)	29420 (0)	5254 (−2)	4386 (−3)	0.23	0.19
2017–18	37893 (8)	30142 (2)	5879 (12)	4677 (7)	0.22	0.18
2018–19	43397 (15)	34615 (15)	6207 (6)	4951 (6)	0.23	0.18
2019–20	40860 (−6)	29724 (−14)	5761 (−7)	4191 (−15)	0.20	0.15
2020–21	38712 (−5)	30797 (4)	5212 (−10)	4146 (−1)	0.20	0.16

Source: As mentioned in the text.

is only about 1.17% of the global space economy, estimated to be USD 447 billion in that year.[11]

The U.S.A. perhaps is the only country where the official statistical agency, namely the Bureau of Economic Analysis of the Ministry of Commerce, estimates the size and structure of the U.S. Space economy (Highfill, Jouard and Franks, 2022)[12] [9]. On average, the Indian space economy (unadjusted) is only about 2.79% of its U.S. counterpart. Further, the U.S. space economy accounted for about 0.60% of the US GDP, while the Indian space economy accounted for about 0.23% of India's G.D.P. in 2019.

5.2. Structure

India's space economy, like the U.K. and the U.S., is dominated by the space applications segment (Table 14). As noted before, within the space applications segment, the most important one is the D.T.H., and within space operations, it is the development of space technology. Apart from D.T.H., the next important segment is the telecom industry, where space technology is used *to provide* location services. Space manufacturing is the smallest of the three. However, with the opening of the space sector to private sector participation, the size of the space manufacturing sector is bound to increase in the years to come. A clear indication of this is seen in the emergence of a number of space start-ups, as most of these are in the manufacture of satellites and subsystems of satellites (see Table 11 above). There has been considerable stability in the shares of the three segments over the years.

5.3. Productivity of space investments

Hitherto, the space economy of India, has been almost entirely created and nurtured by the central government. According to the Department of Science and Technology (2020), the R&D budget of the space sector in 2017–18 accounted for 8.8% of India's Gross Expenditure on R&D or 19% of the R&D budgets of all the 12 scientific agencies in the country. Given the sizeable amounts spent on a high technology sector like space research in a late industrialising and developing country, it will be interesting to measure the productivity of space research in some precise terms. This could be accomplished in two ways. The first is to undertake a detailed survey of all those activities which depend on space technology, quantify their size in terms of output and relate the output to total investments by the government in the space economy. This will provide us with a sort of productivity of space investment in terms of a single number. A movement in the level of this number will signify the productivity of investment: an increase in the ratio will mean a decline in productivity and vice versa. The second is to identify those specific sectors affected by space technology. Based on a literature survey of space-related studies, which included 77 publications covering the period from 1972 to 2018, OCED (2019) [25] has identified a number of sectors where space technology has an impact. The identified sectors or areas are the overall economy at the national level through its contribution to GDP; agriculture; health; transport and urban planning; education; environmental management; climate monitoring and meteorology; energy; telecommunications; disaster management; finance and insurance; manufacturing, mining and construction; high tech industries; defence and security; tourism and leisure; research and development and science; data analytics and location-based services; other generic services. In this paper, we have adopted the former way of measuring the productivity of space investments by relating the amount of space investment to the value of the output created through that investment, as estimated by us and presented in Table 13 above. Based on this line of reasoning, we define two estimates of productivity, and these are explained as follows:

- Estimate 1 = Ratio of the size of the space economy to space budget
- Estimate 2 = Ratio of the adjusted size of the space economy to space budget

Where

Adjusted size of the space economy = Total size of the space economy minus those aspects of the space economy that contain a portion of the space budget. The space budget is included in the components of space operations, such as space technology, space sciences, and other space applications. This netting out of the space budget is done to avoid double counting of the space budget in both input and output.

In other words, the two estimates are nothing but the capital-output ratios, and it measures the amount of space budget required to produce one unit of space output. An increase in this ratio implies a decline in the productivity of that investment. There are, of course, two important caveats. First, we assume

[11] See Ref. [45]. But as seen Table 3 above, according to the think tank STPI the estimates of global space economy by various agencies including spacefoundation.org as over estimates by as much as 98%.

[12] Estimates of the US space economy was initially released on December 17, 2020 and it covered the period 2012–18. Subsequently an updated and revised estimate of the U S economy, 2012–2019 was published on January 26, 2022. The exercise is a one-time project and has estimates of gross output, value added, employment and compensation (wages) for the U S space economy. See Highfill, Jouard and Feranks, 2022) [9] for the details.

Table 14
Structure of India's space economy (Values are in Rs. Crores at current prices).

Year	Space manufacturing	Space operations[a]	Space applications[b]	Total size of space economy
2011–12	1185.39 (5.24)[c]	5296.67 (23.41)	16142 (71.35)	22624 (100)
2012–13	1394.85 (5.58)	5839.29 (23.36)	17761 (71.06)	24995 (100)
2013–14	1821.03 (6.62)	6392.79 (23.25)	19283 (70.13)	27497 (100)
2014–15	3348.90 (10.31)	7251.55 (22.33)	21871 (67.35)	32471 (100)
2015–16	3023.51 (8.61)	8264.76 (23.52)	23845 (67.87)	35133 (100)
2016–17	3161.05 (8.97)	8761.65 (24.86)	23318 (66.17)	35241 (100)
2017–18	2926.94 (7.72)	8733.42 (23.05)	26232 (69.23)	37893 (100)
2018–19	1904.40 (4.39)	10225.72 (23.56)	31267 (72.05)	43397 (100)
2019–20	2590.80 (6.34)	11784.19 (28.84)	26485 (64.82)	40860 (100)
2020–21	3433.33 (8.87)	8217.82 (21.23)	27061 (69.90)	38712 (100)

Source: As mentioned in the text.
Notes:
[a] Space operations include space technology + space sciences + ISTRAC + MCF + NRSA/NRSC + revenue of Antrix.
[b] Space Applications includes 5% of the annual revenue of telecom mobile services + revenue generated from D.T.H. industry + Other Space applications + INSAT.
[c] Figures in parentheses indicate percentage share of the total.

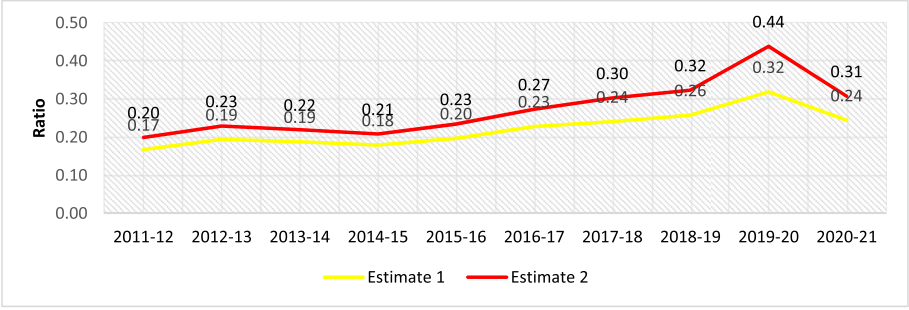


Fig. 6. Productivity of space budget in India, 2011–12 through 2020–21. Source: As mentioned in the text.

Table 15
Framework for analysing the impact of space investments.

Category of impact	In the space sector	In other sectors
New Jobs	The workforce in the space sector	Employment locally, regionally serving the space sector workforce (e.g. local shops, industries). Employment in companies, organisations, using space-related products or services to create new products or services (e.g. imagery in geospatial equipment, satellite signals in navigation equipment).
New Revenues	Revenues from new services	Revenues coming from new services based on space-based elements (telecommunications, navigation, geospatial services)
Efficiency	Increased competitiveness of some firms	Productivity gains are achieved by improving space assets users' production and distribution. Cost savings
Cost avoidance	–	Reduced damage to properties and lives
Social inclusion	–	Satellite communications infrastructure projects contribute to addressing the problem of social exclusion by improving accessibility

Source: OECD (2007).

zero time lags between investment and output as it is difficult to arrive at an acceptable time lag between the incurring of an investment and the output created with that investment. However, using this line of reasoning, it must be stressed that the increasing capital-output ratio in the case of the space economy in India need not be taken as a sign of declining productivity of investments, as some of the investments made in the recent past are for developing complex technologies like cryogenic engines, semi-cryogenic rockets, human spaceflight capability, and so on, and remote sensing which typically have a long gestation period with high failure rates. Second, as a result of the above, the capital-output ratio is a very crude indicator of the productivity of investment in space technology as space technologies are highly complex and have a long gestation period.

Of the two estimates, Estimate 2 is a more meaningful measure of the productivity of space investments as it avoids duplication of data, as mentioned above. The two estimates of productivity are presented in Fig. 6. It shows a steady increase in the capital-output ratios signifying that the productivity of public investments in creating a space economy has been decreasing. The ratio has, of course, fallen in 2020–21, essentially due to an increase in space output, even when the space investment too has declined in that year. Given the two caveats stated above, we do not draw any firm conclusions about the direction of space investments' efficiency.

Although crude, the exercise draws our attention to the need for a detailed study on the impact of space investments on the rest of the economy. Of course, this is a complicated exercise as we have

both first-order (or direct) and second-order (indirect) impacts, see Table 15. Our discussions have presented evidence of two of the first-order impacts: workforce in the space sector (Fig. 2) and revenue from new services such as D.T.H. (Table 7).

6. Conclusions

The space economy of the country is in a state of transition driven by national policies, maturity of space applications and space capabilities diffusion to the private sector. At least four distinct aspects of the space economy need to be emphasised. First, there is a worldwide attempt to delineate the contours of this intensive high-technology industry. Even in the West, statistical agencies are still proceeding on an iterative basis to estimate its size as there is no real consensus as to what constitutes a spacefaring nation's space economy. This is because the space economy spans both manufacturing and services, and in several industries, space technology enters along with other technologies. So separating space technology from other technologies used is fraught with practical difficulties. The case of the telecommunications services industry is a case in point. Second, government budgets play an important role in creating and nurturing the space industry. Although there are not any econometric exercises, it is generally believed that public investments in space technology crowd in private sector investments. Third, space manufacturing and launching technology have evolved so much that it is possible for even small and medium enterprises to enter this industry that hitherto has been the exclusive preserve of large public sector entities such as NASA or ISRO. Fourth, a combined effect of especially the latter two issues has led to large scale entry of private sector enterprises into the space economy.

It is pertinent to point out that public investment in the space is essential to space economy and industry as seen through the top rank by the U.S.A. and a rapid rise of China as a global space technology leader by number of launches and across board technology development through its second largest space budget amongst all space-faring nations.

India's space economy has evolved considerably and now accounts for an average of about 0.23% of her G.D.P. The space applications industry still dominates the sector, and the two most important sub-sectors within the space applications industry are the satellite telecommunications and the D.T.H. industries. Of these, the recent growth of online streaming is likely to challenge the development of the D.T.H. industry. We have also noticed a decline in the state budget for space-related activities, leading to a reduction in the size of the economy in the last two years. The capital-output ratio in the space industry has increased over time, which could either imply a decrease in the productivity of investment as well as investment in the development of next generation high-cost, long-gestation period technologies. However, given the long gestation period and earlier policies that sought to restrict the use of space technology, a rise in the capital-output ratio was inevitable. The slow growth of remote sensing economic return is driven by international efforts to provide moderate resolution data with free and open licencing. The recent spate of public policies directed at the space sector is very likely to enlarge the size of this sector through enhanced private investments, improved integration with the global private space industry and seize new opportunities in space service sector through a larger deployment of the private workforce.

Again, we reiterate the large role of space technology for security, governance and transparency, environment protection and disaster management, which are critical to sustainability and quality of life and would generally be missed out in assessments of the space economy. However, conceptual framework development for indirect impacts and continued monitoring of the Indian space economy requires much more attention as the Indian economy grows and relies more on space technology and its benefits.

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Data availability

Data will be made available on request.

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